

DAMAGE ASSESSMENT

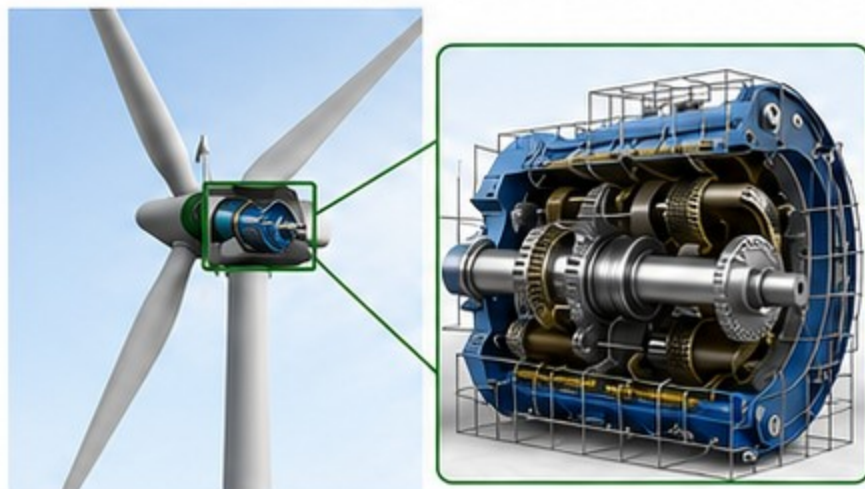
From Wind Turbine Monitoring to Entropy–Damage Correlation

Patent No.: **US 2020/0382022 A1**

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1. APPLICATION: WIND TURBINE

Wind turbine drivetrain
(bearings & gearbox)

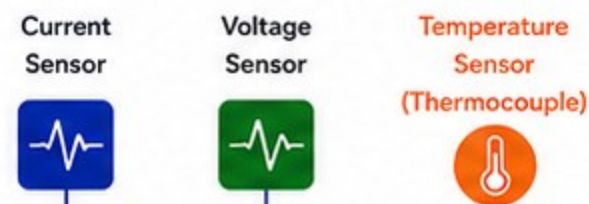


OPERATING CONDITIONS



2. ENTROPY-BASED DAMAGE ASSESSMENT SYSTEM

SENSING & DATA ACQUISITION



Data Acquisition System
(DAQ)

Σ Entropy Generation
Calculation

Damage Assessment
(Using Entropy)

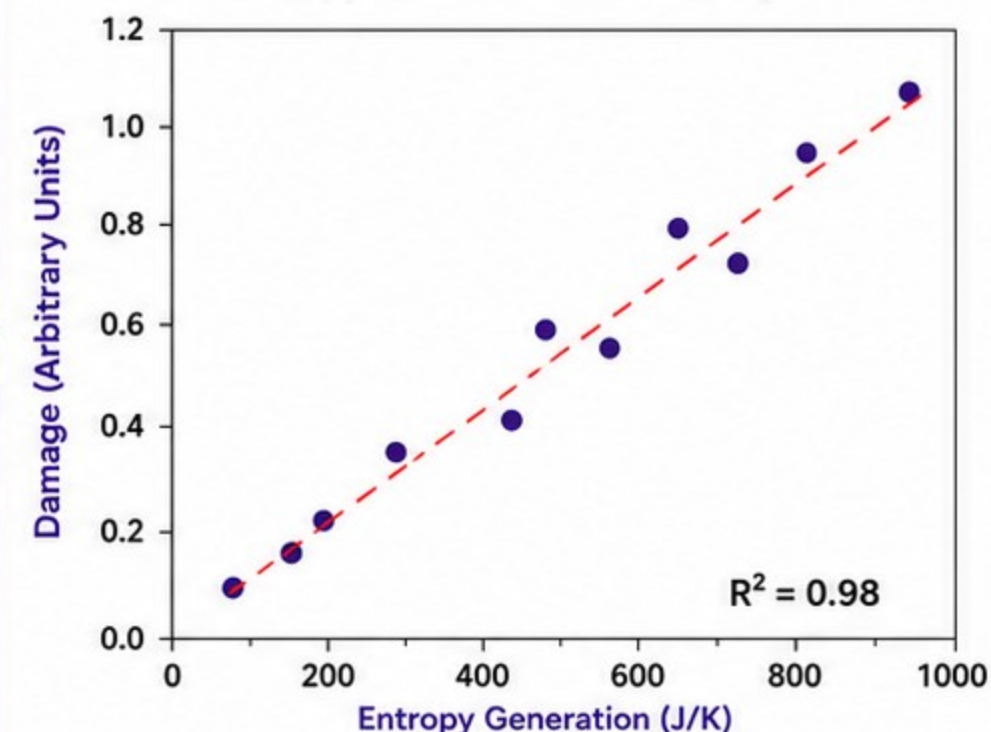
WIRELESS COMMUNICATION



Computer / Dashboard

3. ENTROPY-DAMAGE CORRELATION

Linear Relationship Between
Entropy Generation and Damage



Higher entropy generation indicates
higher damage in the tribological system.

OVERALL WORKFLOW



Key Idea: Entropy generation from easily measurable signals (current, voltage, temperature) provides a physics-informed indicator of damage in tribological systems.